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[This question paper contains 16 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 4432

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Unique Paper Code : 32347507

Name of the Paper : Data Analysis and
Visualization

Name of the Course : B.Sc. (Hons.) Computer
Science

Semester : V

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Question No. 1 is compulsory.
3. Attempt any **four** questions out of Q.2 to Q.7.
4. Parts of a question must be answered together.

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1. (a) Provide code to create a time-series with two index labels- 2011/9/01 and 2011/9/02. Assign random values. (2)

(b) What will be the output of the following codes?

(i) (2)

```
import numpy as np
arr = np.array([[1,2,3,4,5],[6,7,8,9,10]])
print(arr[1,-1], arr[-1:])
```

(ii) (2)

```
List = [str[::-1] for str in ('happy','go','lucky')]
print(List)
```

(c) Reshape the following array to dimension (2,6)

[[3,4,5,6], [7,8,9,10], [11,12,13,14]] (2)

(d) Python is a strongly "typed" language. Comment. (2)

- (e) Give the output of the following code and identify the role of *is_unique* attribute in the code. (2)

```
import pandas as pd
series = pd.Series([4,5,1,2,3,3,4,5,6])
print(series)
print("Is Unique: ",series.is_unique)
```

- (f) Differentiate between mutable and immutable objects. (2)
- (g) Write a program to create the given dataframe. (3)

	id	value
0	1	a
1	1	a
2	2	b
3	3	None
4	3	a
5	4	a
6	4	None
7	4	b

Further, split it into groups and count unique values of 'value' column.

(h) Provide the output of the following code : (3)

```
from datetime import datetime, date, time
dt = datetime(2011, 10, 29, 20, 30, 21)
dt2 = datetime(2011, 11, 15, 22, 30)
delta = dt2 - dt

print(delta)
print(type(delta))
print(dt.replace(minute=0, second=0))
```

(i) Consider the given dataframe *df* containing data of students admitted in the college. (3)

Id	Name	Age	Section	City	Gender	Marks
S0	Anit	10	A	Gurgaon	M	60
S1	Alka	22	B	Delhi	F	80
S2	Sid	13	C	Mumbai	M	60
S3	Ruhi	21	B	Delhi	F	55
S4	Nehu	12	B	Mumbai	F	60
S5	Geet	11	A	Delhi	F	56
S6	Om	17	A	Mumbai	M	45

Set the first column '*Id*' as the row index of the given dataframe *df*. Create a pivot table of *df* to display the total number of admissions as per '*Section*' and '*Gender*'.

(j) (i) Provide the output of the following code :

(4)

```
df = pd.DataFrame({
    'a':np.arange(1,7),
    'b':np.arange(7,13),
    'c':np.arange(12,18),
    'd':np.arange(17,23),
    'e':np.arange(23,29),
    'f':np.arange(29,35)},
    columns=['a', 'b', 'c', 'd', 'e', 'f'],
    index=['Svaksh', 'Sarah', 'Svaraj', 'Rivika',
    'Rahul', 'Geet'])
print(df)
df.iloc[2:4,[1,2]]=np.NaN
print(df)

mapping = {'a': 'red', 'b': 'red', 'c': 'blue', 'd':
'blue', 'e': 'red', 'f': 'orange'}
```

(ii) Using the above dataframe, group *df* by mapping and find the sum.

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- (k) Consider the following dataset to perform the following operations : (4)

	Age	Section	City	Gender	Favourite_color
0	10	A	Gurgaon	M	red
1	22	B	Delhi	F	NaN
2	13	C	Mumbai	F	yellow
3	21	B	Delhi	M	NaN
4	12	B	Mumbai	M	black
5	11	A	Delhi	M	green
6	17	A	Mumbai	F	red

- (i) Find all the rows where Age is greater than or equal to 12 and the Gender is male.
- (ii) If Age is greater than 20, then use the loc function to update Section with "S" and City with Pune.
- (iii) Select rows 1 to 2 with columns 2 to 3

using iloc.

(L) What is the output of the following code : (4)

```
import pandas as pd
fruits=['apple','orange','apple','apple']*2
N=len(fruits)

print(N)
df=pd.DataFrame({'fruit':fruits,'basket_ID':
np.arange(N),'count' :
np.random.randint(3,15,size=N),'weight':
np.random.uniform(0,4, size=N)})
print(df)
Fruit_cat=df['fruit'].astype('category')
print(Fruit_cat)
print(df.dtypes)
```

2. (a) Differentiate between :

(i) qcut and cut methods

(ii) Pandas.merge and pandas.concat (2)

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- (b) Consider the following numeric grades (out of 4).
Formulate bins for the given grades as per the
following condition : (3)

Below 2.5	Very bad
Between 2.5 to 3	Bad
Between 3 to 3.25	Average
Between 3.25 to 3.5	Good
Between 3.5 to 3.75	Very good
Between 3.75 to 4	Excellent

- (c) Given the following dataframe, provide the output
for the following commands : (5)

	ord_no	purch_amt	ord_date	customer_id
0	NaN	NaN	NaN	NaN
1	NaN	270.65	2012-09-10	3001.0
2	70002.0	65.26	NaN	3001.0
3	NaN	NaN	NaN	NaN
4	NaN	948.50	2012-09-10	3002.0
5	70005.0	2400.60	2012-07-27	3001.0
6	NaN	5760.00	2012-09-10	3001.0
7	70010.0	1983.43	2012-10-10	3004.0
8	70003.0	2480.40	2012-10-10	3003.0
9	70012.0	250.45	2012-06-27	3002.0
10	NaN	75.29	2012-08-17	3001.0
11	NaN	NaN	NaN	NaN

- (i) `df.dropna(thresh=2)`
- (ii) `df.dropna(how='all')`
- (iii) `df.dropna(how='all', axis=1)`
- (iv) `df.isnull()`
- (v) `df.isnull().values.any()`

3. (a) Write the code to read each row of a given csv file. Skip the header of the file while reading. Also print the number of rows and the field names.

(6)

- (b) (i) Differentiate between `ffill` and `bfill`. (4)

- (ii) Provide the output of the given code :

```
import pandas as pd
obj3 = pd.Series(['blue', 'purple', 'yellow'],
                 index=[0, 2, 4])
print(obj3.reindex(range(6), method='ffill'))
print(obj3.reindex(range(6), method='bfill'))
```

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4. (a) Consider following dataframe.

ord_no	purch_amt	ord_date	customer_id	salesman_id
70009.0	890.00	2012-09-11	3004.0	5001
70002.0	270.65	2012-09-10	3001.0	5006
70007.0	65.26	2012-09-11	3001.0	5005
70008.0	78.00	2012-09-10	3002.0	5003
70006.0	948.50	2012-09-17	3002.0	5002
70005.0	2400.60	2012-07-27	3001.0	5001
70004.0	5760.00	2012-09-10	3001.0	5003
70010.0	1983.43	2012-10-10	3004.0	5006
70003.0	2480.40	2012-10-10	3003.0	5005
70012.0	250.45	2012-06-27	3002.0	5002
70034.0	75.29	2012-08-17	3001.0	5004
70022.0	56.90	2012-06-27	3003.0	5005

With respect to the above dataframe, write the code for the following :

- (i) Group the data on the column *ord_date* and calculate the total purchase amount *purch_amt* year wise and month wise.

(2)

- (ii) Group the data on the column *customer_id* and create a list of order date *ord_date* for each group. (2)
- (iii) Group on the columns *customer_id*, *salesman_id* and then sort sum of *purch_amt* within the groups. (2)
- (b) (i) Write a generator function to print Fibonacci numbers. (4)
- (ii) What is the output of the following code :

```
def simpleGeneratorFunc():  
    yield 1  
    yield 2  
x = simpleGeneratorFunc()  
  
print(next(x))  
print(next(x))
```

5. (a) Give output of the following code. Justify your answer. (2)

```
var=(1, 2, (3,4))  
var[1]='geet'  
print(var)
```

- (b) Write the code to merge the two given datasets using key1, key2. (4)

data1:

	key1	key2	P	Q
0	K0	K0	P0	Q0
1	K0	K1	P1	Q1
2	K1	K0	P2	Q2
3	K2	K1	P3	Q3

data2:

	key1	key2	R	S
0	K0	K0	R0	S0
1	K1	K0	R1	S1
2	K1	K0	R2	S2
3	K2	K0	R3	S3

- (c) Write the code to split the given dataset into groups based on *customer_id* and create a list of order date *ord_date* for each group.

(4)

	ord no	purch amt	ord_date	customer_id
0	70009.0	890.00	2012-09-11	3004.0
1	70002.0	270.65	2012-09-10	3001.0
2	70007.0	65.26	2012-09-11	3001.0
3	70008.0	78.00	2012-09-10	3002.0
4	70006.0	948.50	2012-09-17	3002.0
5	70005.0	2400.60	2012-07-27	3001.0
6	70004.0	5760.00	2012-09-10	3001.0
7	70010.0	1983.43	2012-10-10	3004.0
8	70003.0	2480.40	2012-10-10	3003.0
9	70012.0	250.45	2012-06-27	3002.0
10	70034.0	75.29	2012-08-17	3001.0
11	70022.0	56.90	2012-06-27	3003.0

6. (a) Create a Timeseries Dataframe with date range 01-02-2022 to 30-02-2022 with 1 min frequency interval. The dataframe has two columns populated with random values.

(3)

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(b) Identify the need to resample Timeseries data.

(2)

(c) Consider following dataset.

Datetime	value1	value2	value3
2020-01.01 00:00:00	2	92	56
2020-01.01 00:01:00	9	78	80
2020-01.01 00:02:00	69	83	43
2020-01.01 00:03:00	47	62	45
2020-01.01 00:04:00	47	90	13
...	...	...	...
2020-02.27 23:56:00	73	81	35
2020-02.27 23:57:00	20	66	58
2020-02.27 23:58:00	42	16	48
2020-02.27 23:59:00	32	40	19
2020-02.28 00:00:00	37	63	95
83521 rows x 3 columns			

(i) Resample for 10min with sum function for *value1*, mean for *value2* and max for *value 3*.

(3)

- (ii) Downsample data to 30s. (2)

7. (a) Create a DataFrame of 8 rows and 8 columns containing random integers in the range of 1 to 10. Compute the correlation of each row with the preceding row. (2)

- (b) Consider the following table that lists the last week Delhi's AQI.

AQI	Date
67	2/10/2022
79	3/10/2022
80	4/10/2022
90	5/10/2022
99	6/10/2022
110	7/10/2022
112	8/10/2022
140	9/10/2022
165	10/10/2022
178	11/10/2022

- (i) Plot a line graph showing AQI (Air Quality Index) against date with line colour as red, line width as "4pixels" and dashed line style. (4)

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- (ii) Add title "Delhi AQI for last ten days". (1)
- (iii) Set label for x-axis "Date" and y-axis "AQI". (1)
- (iv) Show grids in the background. (1)
- (v) Set marker as '*'. (1)